

SYNERGEN G2D

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Substance key: 000000285791

Revision Date: 24.08.2022

Version : 1 - 0 / T

Date of printing : 05.04.2026

1. PRODUCT AND COMPANY IDENTIFICATION

Trade name

SYNERGEN G2D

Material number: 178546

Substance name: Fatty alkyl amine derivative

Use of the Substance/Mixture

Industry sector : Personal Care
Crop Protection
Type of use : Surfactant

Supplier name/address/telephone no.:

Clariant (Australia) Pty Ltd.
ACN 069 435 552 ABN 30 069 435 552
c/o Waterman Chadstone, Lvl 2, UL/40, 1341 Dandenong Road
Chadstone, Victoria 3148, Australia
Telephone : +61 3 8562 3300

Information about the substance/preparation

Telephone: +61 3 8562 3300
Facsimile: +61 3 8562 3333
E-mail address: SDS.SEAP@clariant.com

Emergency telephone number: +65 3158 1074

2. HAZARDS IDENTIFICATION

GHS Classification

Skin corrosion/irritation : Category 2

Serious eye damage/eye irritation : Category 1

GHS label elements

Hazard pictograms :



Signal word : Danger

Hazard statements : H315 Causes skin irritation.
H318 Causes serious eye damage.

Precautionary statements : **Prevention:**
P264 Wash skin thoroughly after handling.
P280 Wear protective gloves/ eye protection/ face protection.
Response:

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P302 + P352 IF ON SKIN: Wash with plenty of soap and water.
P305 + P351 + P338 + P310 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Immediately call a POISON CENTER or doctor/ physician.

P332 + P313 If skin irritation occurs: Get medical advice/ attention.

P362 Take off contaminated clothing and wash before reuse.

Other hazards which do not result in classification

Irritating to skin.

Risk of serious damage to eyes.

3. COMPOSITION/INFORMATION ON INGREDIENTS

Substance / Mixture : Mixture

Components

Chemical name	CAS-No.	Concentration (% w/w)
Betaines, C12-14-alkyldimethyl	66455-29-6	>= 20 -< 30
Sodium chloride	7647-14-5	>= 1 -< 10

4. FIRST AID MEASURES

- General advice : Remove/ Take off immediately all contaminated clothing.
Get medical advice/ attention if you feel unwell.
- If inhaled : If inhaled, remove to fresh air.
Get medical advice/ attention.
If breathing has stopped, apply artificial respiration.
Seek medical advice.
- In case of skin contact : Wash off immediately with plenty of water.
Consult a physician.
- In case of eye contact : Rinse immediately with plenty of water, also under the eyelids,
for at least 15 minutes.
Consult a physician.
- If swallowed : Do not induce vomiting. If conscious, give 2 glasses of water.
Get immediate medical attention.
- Most important symptoms and effects, both acute and delayed : None known.

5. FIREFIGHTING MEASURES

Suitable extinguishing media : all

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- Unsuitable extinguishing media : No restrictions
- Specific extinguishing methods : Cool closed containers exposed to fire with water spray.
- Special protective equipment for firefighters : Full protective suit
Self-contained breathing apparatus

6. ACCIDENTAL RELEASE MEASURES

- Personal precautions, protective equipment and emergency procedures : Prevent further leakage or spillage.
Collect spillage.
Flush away traces with water.
Do not empty into drains.
Do not dispose of waste into sewer.
- Methods and materials for containment and cleaning up : Soak up with inert absorbent material (e.g. sand, silica gel, acid binder, universal binder, sawdust).
Keep in suitable, closed containers for disposal.

7. HANDLING AND STORAGE

- Advice on protection against fire and explosion : No special measures necessary.
- Further information on storage conditions : - sensitive to frost - In case of the product becoming opaque, thickening or being frozen due to the effects of cold, allow to thaw slowly at room temperature. Stir briefly before use.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION**Components with workplace control parameters**

Contains no substances with occupational exposure limit values.

Personal protective equipment

- Respiratory protection : Use respiratory protection in case of insufficient exhaust ventilation or prolonged exposure
Full mask, filter A
Full mask to standard DIN EN 136
Filter class 2
The use of filter apparatus presupposes that the environment atmosphere contains at least 17% oxygen by volume, and does not exceed the maximum gas concentration, usually 0.5% by volume. Relevant guidelines to be considered include EN 136/141/143/371/372 as well as other national regulations.
Applicable national Regulations must be observed. Take note of the limitations regarding wear-time, in conjunction with the Regulations for the use of Respiratory Protective Equipment.

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Hand protection	
Remarks	: Protective gloves
Eye protection	: Depending on the risk, wear sufficient eye protection (safety glasses with side protection or goggles, and if necessary, face shield.) For details, see the BG-rules number 192.
Skin and body protection	: Wear suitable protective equipment.
Protective measures	: Avoid contact with skin and eyes.

9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance	: Liquid
Colour	: light yellow
Odour	: No information available.
pH	: 7 - 8.5 Concentration: 1 %
Boiling point	: approx. 100 °C
Flash point	: No flash point - Measure made up to the boiling point.
Flammability (solid, gas)	: Not applicable
Self-ignition	: Not applicable
Upper explosion limit / upper flammability limit	: not tested.
Lower explosion limit / Lower flammability limit	: not tested.
Relative vapour density	: no data available
Density	: approximately 1.03 g/cm ³
Solubility(ies)	
Water solubility	: miscible
Auto-ignition temperature	: no data available
Viscosity	
Viscosity, dynamic	: < 30 mPa.s

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Viscosity, kinematic	: no data available
Metal corrosion rate	: Not applicable
Particle size	: Not applicable

10. STABILITY AND REACTIVITY

Reactivity	: No dangerous reaction known under conditions of normal use.
Possibility of hazardous reactions	: No dangerous reaction known under conditions of normal use.
Conditions to avoid	: None known.
Incompatible materials	: none
Hazardous decomposition products	: No hazardous decomposition products if stored and handled as prescribed

11. TOXICOLOGICAL INFORMATION**Acute toxicity****Product:**

Acute oral toxicity	: Remarks: not tested.
	Acute toxicity estimate: > 5,000 mg/kg
	Method: Calculation method

Acute inhalation toxicity	: Remarks: not tested.
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Acute dermal toxicity	: Remarks: not tested.
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Components:**Sodium chloride:**

Acute oral toxicity	: LD50 (Rat, male): 3,550 mg/kg
	Method: No information available.
	GLP: no data available

Skin corrosion/irritation**Product:**

Remarks	: not tested.
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Components:**Betaines, C12-14-alkyldimethyl:**

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Result : Causes burns.

Sodium chloride:

Species : Rabbit
Method : No information available.
Result : No skin irritation
GLP : no

Serious eye damage/eye irritation**Product:**

Remarks : not tested.

Components:**Sodium chloride:**

Species : Rabbit
Result : No skin irritation
Method : OECD Test Guideline 405
GLP : no

Respiratory or skin sensitisation**Product:**

Remarks : not tested.

Components:**Sodium chloride:**

Test Type : Local lymph node assay (LLNA)
Exposure routes : Dermal
Species : Mouse
Method : Other
Result : Not a skin sensitizer.
GLP : no data available

Germ cell mutagenicity**Components:****Sodium chloride:**

Genotoxicity in vitro : Test Type: In vitro mammalian cell gene mutation test
Test system: mouse lymphoma cells
Concentration: 0,25 - 5,85 mg/ml
Metabolic activation: without
Method: Other
Result: positive
GLP: no data available

Test Type: Ames test
Test system: Salmonella typhimurium
Metabolic activation: with and without metabolic activation

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Method: OECD Test Guideline 471

Result: negative

GLP: no data available

Genotoxicity in vivo : Test Type: Chromosome Aberration Test
Species: Rat (female)
Strain: Long-Evans
Cell type: Bone marrow cells
Application Route: Intraperitoneal injection
Exposure time: 3-6-12-18-24 h
Dose: 10 - 20 - 40 mmole/kg
Method: OECD Test Guideline 475
Result: positive
GLP: no data available

Germ cell mutagenicity - Assessment : Mutagenicity not fully evaluated.

Carcinogenicity**Components:****Sodium chloride:**

Species : Rat, male
Application Route : oral (feed)
Exposure time : 2 a
Dose : 4% NaCl and 2% KCl+2% NaCl
Control Group : yes
Frequency of Treatment : daily
Method : 2,533 mg/kg bw/day
GLP : OECD Test Guideline 453
GLP : no data available

Carcinogenicity - Assessment : Animal testing did not show any carcinogenic effects.

Reproductive toxicity**Components:****Sodium chloride:**

Effects on foetal development : Remarks: The study is not necessary from a scientific perspective.

Reproductive toxicity - Assessment : No reproductive toxicity to be expected.
No teratogenic effects to be expected.

STOT - single exposure**Product:**

Remarks : no data available

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Components:**Sodium chloride:**

Assessment : The substance or mixture is not classified as specific target organ toxicant, single exposure.

STOT - repeated exposure**Product:**

Remarks : no data available

Components:**Sodium chloride:**

Assessment : The substance or mixture is not classified as specific target organ toxicant, repeated exposure.

Repeated dose toxicity**Product:**

Remarks : not tested.

Components:**Sodium chloride:**

Species : Rat, male
LOEL : 2533 mg/kg bw/day
Application Route : oral (feed)
Exposure time : 2 a
Number of exposures : daily
Dose : 4% NaCl and 2% KCl+2% NaCl
Control Group : yes
Method : OECD Test Guideline 453
GLP : no data available

Application Route : Inhalation
Method : Repeated dose toxicity
Remarks : The study is not necessary from a scientific perspective.

Aspiration toxicity**Product:**

no data available

Components:**Sodium chloride:**

No aspiration toxicity classification

Further information**Product:**

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Remarks : The product has not been tested. The information is derived from the properties of the individual components.

12. ECOLOGICAL INFORMATION**Ecotoxicity****Product:**

Toxicity to fish : Remarks: not tested.

Toxicity to daphnia and other aquatic invertebrates : Remarks: not tested.

Toxicity to algae/aquatic plants : Remarks: not tested.

Toxicity to microorganisms : Remarks: no data available

Components:**Sodium chloride:**

Toxicity to fish : LC50 (Lepomis macrochirus (Bluegill sunfish)): 5,840 mg/l
End point: mortality
Exposure time: 96 h
Test Type: flow-through test
Analytical monitoring: no
Method: Other
GLP: no

Toxicity to daphnia and other aquatic invertebrates : LC50 (Daphnia magna (Water flea)): 4,136 mg/l
End point: Immobilization
Exposure time: 48 h
Test Type: static test
Analytical monitoring: yes
Method: OECD Test Guideline 202
GLP: no
Remarks: Acute Immobilisation Test

Toxicity to algae/aquatic plants : ErC50 (Nitzschia sp.): 2,430 mg/l
End point: Growth rate
Exposure time: 120 h
Test Type: static test
Analytical monitoring: no
Method: OECD Test Guideline 201
GLP: no

IC50 (Lemna minor (common duckweed)): 6,870 mg/l
End point: Growth rate
Exposure time: 96 h
Test Type: static test
Analytical monitoring: no
Method: OECD Test Guideline 221

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GLP: no data available

Remarks: The details of the toxic effect relate to the nominal concentration.

Toxicity to fish (Chronic toxicity)

: NOEC (Pimephales promelas (fathead minnow)): 252 mg/l
End point: mortality
Exposure time: 33 d
Test Type: flow-through test
Analytical monitoring: yes
Method: OECD Test Guideline 210
GLP: no

LOEC (Pimephales promelas (fathead minnow)): 352 mg/l
End point: mortality
Exposure time: 33 d
Test Type: flow-through test
Analytical monitoring: yes
Method: OECD Test Guideline 210
GLP: no

Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity)

: NOEC (Daphnia pulex (Water flea)): 314 mg/l
Exposure time: 21 d
Test Type: semi-static test
Analytical monitoring: yes
Method: OECD Test Guideline 211
GLP: no

LOEC (Daphnia pulex (Water flea)): 441 mg/l
Exposure time: 21 d
Test Type: semi-static test
Analytical monitoring: yes
Method: OECD Test Guideline 211
GLP: no

Toxicity to microorganisms

: EC10 (activated sludge): <= 35,000 mg/l
Test Type: aquatic
Analytical monitoring: no
Method: Other
GLP: no

Test Type: Soil
Remarks: Not applicable

Toxicity to soil dwelling organisms

: Test Type: artificial soil
NOEC (Eisenia fetida (earthworms)): 3,507 mg/kg
Exposure time: 70 d
End point: mortality
Method: OECD Test Guideline 222
GLP: no

Test Type: artificial soil
LOEC (Eisenia fetida (earthworms)): 4,675 mg/kg
Exposure time: 70 d
End point: mortality

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Method: OECD Test Guideline 222

GLP: no

Test Type: artificial soil

LOEC (Eisenia fetida (earthworms)): 1,169 mg/kg

Exposure time: 70 d

End point: Reproduction

Method: OECD Test Guideline 222

GLP: no

Plant toxicity

: IC50: 500.8 mg/kg

Exposure time: 7 d

End point: emergence

Species: Lepidium sativum (cress)

Analytical monitoring: no

Method: OECD Guide-line 208

GLP: no

Remarks: The details of the toxic effect relate to the nominal concentration.

NOEC: 360 mg/kg

Exposure time: 7 d

End point: Growth

Species: Lepidium sativum (cress)

Analytical monitoring: no

Method: OECD Guide-line 208

GLP: no

Remarks: The details of the toxic effect relate to the nominal concentration.

LOEC: 640 mg/kg

Exposure time: 7 d

End point: Growth

Species: Lepidium sativum (cress)

Analytical monitoring: no

Method: OECD Guide-line 208

GLP: no

Remarks: The details of the toxic effect relate to the nominal concentration.

IC50: 1,890 mg/kg

Exposure time: 7 d

End point: emergence

Species: Hordeum vulgare

Analytical monitoring: no

Method: OECD Guide-line 208

GLP: no

Remarks: The details of the toxic effect relate to the nominal concentration.

NOEC: 296 mg/kg

Exposure time: 7 d

End point: Growth

Species: Hordeum vulgare

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Analytical monitoring: no
Method: OECD Guide-line 208
GLP: no
Remarks: The details of the toxic effect relate to the nominal concentration.

LOEC: 528 mg/kg
Exposure time: 7 d
End point: Growth
Species: Hordeum vulgare
Analytical monitoring: no
Method: OECD Guide-line 208
GLP: no
Remarks: The details of the toxic effect relate to the nominal concentration.

Toxicity to terrestrial organisms : LD50 (Passer domesticus (house sparrow)): 8,000 mg/kg
Exposure time: 12 h
End point: mortality
Method: Other
GLP: no

Persistence and degradability**Product:**

Biodegradability : Remarks: This property is substance specific and cannot be given for the preparations.

Components:**Sodium chloride:**

Biodegradability : Remarks: Not applicable for inorganic compound.

Physico-chemical removability : Remarks: Not applicable

Stability in water : Remarks: Not applicable

Bioaccumulative potential**Components:****Sodium chloride:**

Partition coefficient: n-octanol/water : Remarks: Not applicable inorganic

Mobility in soil

no data available

Other adverse effects**Product:**

Additional ecological : No ecological testing was carried out on the product. Data are

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information

based on the properties of the individual components

Components:**Sodium chloride:**

Environmental fate and pathways : not available

Results of PBT and vPvB assessment : The substance is inorganic, thus a PBT and vPvB criteria assessment is not applicable according to Annex XIII of Regulation (EC) 1907/2006.

13. DISPOSAL CONSIDERATIONS**Disposal methods**

Waste from residues : Dispose of in accordance with local regulations.

Contaminated packaging : Contaminated packages or drums must be treated as waste, and must be disposed of or treated for reuse/recycling in an installation approved by Environmental Authorities according to local regulations. The wastes generated from such treatment of the packages must be processed in order to avoid environmental contamination.

14. TRANSPORT INFORMATIONIATA not restricted
IMDG not restricted**15. REGULATORY INFORMATION****Safety, health and environmental regulations/legislation specific for the substance or mixture**Hazardous Substance Act : Conditions of restriction for the following entries should be considered:
Not applicable

Emergency Decree on Controlling the Use of Volatile Substances : Not applicable

16. OTHER INFORMATION

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Further information

Other information : Observe national and local legal requirements

Date format : yyyy/mm/dd

Full text of other abbreviations

AIIC - Australian Inventory of Industrial Chemicals; ANTT - National Agency for Transport by Land of Brazil; ASTM - American Society for the Testing of Materials; bw - Body weight; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DSL - Domestic Substances List (Canada); ECx - Concentration associated with x% response; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; ERG - Emergency Response Guide; GHS - Globally Harmonized System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organization; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardization; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; n.o.s. - Not Otherwise Specified; Nch - Chilean Norm; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NOM - Official Mexican Norm; NTP - National Toxicology Program; NZIoC - New Zealand Inventory of Chemicals; OECD - Organization for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; TCSI - Taiwan Chemical Substance Inventory; TDG - Transportation of Dangerous Goods; TECI - Thailand Existing Chemicals Inventory; TSCA - Toxic Substances Control Act (United States); UN - United Nations; UNRTDG - United Nations Recommendations on the Transport of Dangerous Goods; vPvB - Very Persistent and Very Bioaccumulative; WHMIS - Workplace Hazardous Materials Information System

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applicable law. You should obtain and review the applicable Material Safety Data Sheet information before handling any of these products. For additional information, please contact Clariant.

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